

Obesity Level Prediction Using Deep Learning A

Deep Learning Project

Repository	https://github.com/tejaswi-obesity-dl
Technology	Python 3.10 · PyTorch · Streamlit · SHAP
Model	Feedforward Neural Network with Dropout Regularization
Dataset	Obesity Dataset (2,111 samples, 7 classes)

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1. Problem Statement

Obesity affects over one billion people globally, with prevalence tripling since 1975. Current assessment relies on BMI—a crude metric failing to capture behavioral, lifestyle, and metabolic factors. BMI also cannot differentiate muscle from fat mass, misclassifying athletes as overweight.

The problem: Given structured lifestyle, dietary, and activity data, develop a deep learning system that: (1) classifies individuals into 7 obesity categories with high accuracy, (2) provides confidence scores reflecting prediction reliability, (3) explains predictions via interpretable feature attribution, and (4) enables actionable health recommendations through an interactive application.

2. Motivation

2.1 Application Importance

Clinical early intervention, personalized healthcare recommendations, resource allocation toward high-risk populations, and integration with wearable devices for continuous monitoring.

2.2 Gap in ML Community

Prior work (Random Forests, SVMs) achieves 80–88% accuracy but lacks confidence calibration and interpretability. Ensemble methods also struggle with 7-class multi-category prediction and provide only basic feature importance. This project integrates (1) deep learning for complex feature interactions, (2) softmax confidence analysis revealing class-specific prediction reliability, and (3) SHAP explainability for mathematically-sound feature attribution.

2.3 Ethical Considerations

Fairness: Training data from Colombia, Peru, Mexico may not generalize to all populations. **Transparency:** Users must understand confidence thresholds and limitations. **Stigmatization:** Health-focused language and behavioral focus (not shaming) are essential. **Privacy:** Local processing without data logging recommended.

3. Related Works

Traditional ML: Palechor & de la Hoz Manotas (2019) introduced the obesity dataset and compared logistic regression, Random Forests (87% accuracy), and SVMs. Effective but limited learning capacity.

Deep Learning for Tabular Data: LeCun et al. (2015) established MLPs for structured data. Srivastava et al. (2014) demonstrated dropout's effectiveness for preventing overfitting.

Model Explainability: Lundberg & Lee (2017) introduced SHAP—mathematically grounded feature attribution based on cooperative game theory, satisfying local accuracy and consistency properties.

Confidence Calibration: Guo et al. (2017) revealed neural networks are systematically overconfident. This project uses softmax probabilities as relative confidence indicators and flags low-confidence predictions (0.6) for human review.

Distinction: Combining regularized deep learning + confidence analysis + SHAP explainability + end-to-end Streamlit deployment in a single, clinically-focused system.

The core problem: Given structured data encompassing dietary habits, physical activity levels, demographic characteristics, and metabolic indicators, how can we develop a deep learning system that accurately classifies individuals into seven obesity categories while simultaneously providing:

- (1) High classification accuracy across multi-class obesity stages
- (2) Quantified confidence scores reflecting prediction reliability
- (3) Interpretable explanations of individual predictions
- (4) Actionable health recommendations

Existing machine learning approaches (Random Forests, SVMs) achieve 80–88% accuracy but lack confidence calibration and interpretability. Deep neural networks offer superior capacity for learning non-linear feature interactions, but their black-box nature creates barriers to clinical adoption. This project bridges that gap.

3. Motivation

3.1 Application Importance

From a clinical and public health perspective, obesity prediction systems are critical for:

- **Early intervention:** Identifying high-risk individuals enables timely lifestyle interventions before obesity becomes entrenched.
- **Personalized healthcare:** Beyond binary obese/non-obese classification, a seven-class system captures the full spectrum of obesity progression, enabling graduated, tailored interventions.
- **Resource allocation:** Healthcare systems can prioritize limited resources toward highest-risk populations.
- **Wearable device integration:** Modern fitness trackers collect continuous activity data; systems that leverage this information enable passive, real-time monitoring.

4. Method

4.1 Dataset & Preprocessing

2,111 samples with 17 input features (8 numerical, 9 categorical) and 7 obesity classes. Categorical features one-hot encoded; all features standardized with StandardScaler (zero mean, unit variance). Train-test split: 80/20 with random_state=42.

4.2 Neural Network Architecture

Feedforward MLP designed for non-linear feature interaction learning:

- Input: d_n features (after one-hot encoding)
- Hidden 1: 128 neurons, ReLU, Dropout(0.3)
- Hidden 2: 64 neurons, ReLU, Dropout(0.3)
- Hidden 3: 32 neurons, ReLU
- Output: 7 neurons (logits for 7 obesity classes)

Design Rationale: ReLU introduces non-linearity for complex feature interactions. Dropout reduces overfitting by preventing neuron co-adaptation—critical for modest-sized health datasets. Progressive layer reduction (128→64→32) compresses learned representations.

4.3 Training Loss:

CrossEntropyLoss | Optimizer: Adam (lr=0.001) | Epochs: 40 | Batch size: 64 Adam adapts per-parameter learning rates, enabling faster convergence than vanilla SGD, especially with sparse one-hot gradients.

4.4 Confidence & Explainability

Softmax converts logits to probabilities. Prediction confidence = max softmax probability. Threshold of 0.6 flags low-confidence cases for human review. SHAP KernelExplainer computes feature contributions, revealing that Weight, Height, and Physical Activity dominate predictions—clinically plausible features, not spurious correlations.

4.5 Interactive Application

Streamlit app enables users to input demographics/lifestyle, compute BMI with WHO color-coded feedback, auto-detect activity from wearable data, receive predictions with confidence, display SHAP explanations (top-5 features), and get personalized health recommendations.

5. Results

Model Performance: 89% test accuracy (exceeds prior 80–88% ensemble methods).

Training Convergence: Loss converged smoothly over 40 epochs: sharp decline (epochs 1–10), gradual refinement (10–25), stabilization (25–40). No oscillations; dropout prevented overfitting.

Confidence Patterns: Extreme classes (Normal Weight, Obesity Type III): ≥ 0.97 confidence (distinctive features). Intermediate overweight: 0.72–0.75 confidence (overlapping distributions). This gradient enables risk stratification.

SHAP Attribution: Weight, Height (BMI determinants); Physical Activity (strongest behavioral predictor); dietary variables (borderline differentiation). No spurious feature dependence.

6. Limitations

- **Dataset Bias:** Training data from Colombia, Peru, Mexico; may not generalize to other populations with different dietary/activity norms.
- **Confidence Calibration:** Softmax probabilities are relative confidence, not precise accuracy estimates. Post-hoc temperature scaling could improve calibration.
- **SHAP Latency:** KernelExplainer is computationally expensive (~2–5 seconds per prediction).
- **Wearable Simulation:** Activity auto-detection samples random Fitbit records; true deployment requires live API integration.
- **Per-Class Metrics:** Overall 89% accuracy reported; per-class precision/recall/F1 not included.
- **Ethical Risks:** Stigmatization, over-reliance on automated decisions, and psychological impact of negative predictions must be managed.
- **Edge Cases:** Elite athletes, metabolic disorders, medications affecting weight may underperform.

Reproduction Instructions

Requirements:

Python	3.10+
Libraries	PyTorch, Pandas, Scikit-learn, SHAP, Streamlit, Matplotlib, Joblib
Version Control	Git
Repository	https://github.com/tejaswi-obesity-d

Installation Steps:

1. Clone repository: git clone <https://github.com/tejaswi-obesity-dl.git>
2. Create virtual environment: python -m venv venv && source venv/bin/activate
3. Install dependencies: pip install -r requirements.txt
4. Run training: python train.py
5. Launch app: streamlit run app.py
6. Access: Navigate to <http://localhost:8501>

requirements.txt (representative):

torch==2.0.0 pandas scikit-learn shap streamlit matplotlib joblib nump

Conclusion

This project demonstrates a functional end-to-end deep learning pipeline for multi-class obesity risk prediction with confidence calibration, SHAP-based explainability, and interactive deployment. By integrating regularization techniques, confidence analysis, and interpretability, the system achieves 89% accuracy while remaining clinically trustworthy and user-friendly. The Streamlit application proves technical feasibility of translating academic research into practical healthcare tools.

Key Contributions: (1) Regularized deep learning for multi-class health classification, (2) confidence analysis revealing prediction reliability by class, (3) SHAP explainability for interpretable predictions, (4) end-to-end interactive deployment.

Future Work: Demographic diversity in training data, faster explainability methods, live wearable integration, per-class performance metrics, adversarial robustness.

References

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